

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

1st Semester of B. Tech. (CL/EC/EE/ME/IT/CE) Examination

November-2013

MA 101: Engineering Mathematics-I

Date: 25.11.13, Monday

Time: 10:00 a.m. to 01:00 p.m.

Total: 70 Marks

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

Q-1

(3+4)

(a) Find the values of $\text{Log}(1+i)$ and $\log(1+i)$

(b) Using Row Echelon method, find rank of matrix $A = \begin{bmatrix} 8 & 1 & 3 & 6 \\ 0 & 3 & 2 & 2 \\ -8 & -1 & -3 & 4 \end{bmatrix}$

Q-2

(6+4+4)

(a) For which value of k , the system of equations:

$$2x+2y+5z=1, \quad 3x+2y+5z=2k, \quad 3x+4y+10z=k^2$$

have solutions also find complete solution in each case.

(b) Show that infinite series $\sum_{n=0}^{\infty} \left(\frac{\log n}{1000} \right)^n$ is divergent.

(c) Discuss the convergence of infinite series $\sum_{n=0}^{\infty} n! e^{-n}$

OR

Q-2

(5+5+4)

(a) Using Gauss Jordan Method, find inverse of matrix $A = \begin{bmatrix} 8 & 4 & 3 \\ 2 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix}$

(b) Using Gauss Elimination method, find solution of following system of equations
 $3x + 2y + 4z = 7$, $2x + y + z = 4$, $x + 3y + 5z = 2$

(c) Find the Interval of convergence of power series $\sum_{n=0}^{\infty} \left(\frac{n+1}{n+2} \right)^2 x^n$

Q-3

(3+3+4+4)

(a) Find solution of complex equation: $e^{2z} = 1 + i$

(b) Find solution of complex equation: $\cosh z = 0$

(c) Find the roots of $(-16)^{1/4}$

(d) Define the rank of matrix. Find the rank of following matrix by using

Minors (determinant) $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ -2 & -3 & -1 \end{bmatrix}$

OR

Q-3

(4+4+3+3)

(a) Simplify: $\frac{(\cos 5\theta - i \sin 5\theta)^{\frac{2}{5}} \left(\cos \frac{2}{5}\theta + i \sin \frac{2}{5}\theta \right)^5}{(\cos 3\theta + i \sin 3\theta)^{\frac{1}{3}} \left(\cos \frac{2}{3}\theta - i \sin \frac{2}{3}\theta \right)^{18}}$

(b) Show that: $(1 + i\sqrt{3})^{90} + (1 - i\sqrt{3})^{90} = 2^{91}$

(c) Discuss the convergence of infinite series $\sum_{n=2}^{\infty} \ln n$

(d) Write the function $f(z) = z^3 + z + 1$ in the form $f(z) = u(x, y) + iv(x, y)$

SECTION-II

Q-4

(3+4)

- (a) Evaluate: $J = \frac{\partial(u,v)}{\partial(x,y)}$ for $u = x(1-y)$, $v = xy$
- (b) If sum of three positive rational numbers is one then find maximum value of their product.

Q-5

(3+3+4+4)

- (a) Find percentage error in the area of ellipse when an error of 1% made in measuring of semi-major and semi-minor axes.
- (b) If the sides and angles of plane triangle vary in such a way that its circum radius is constant then prove that $\frac{da}{\cos A} + \frac{db}{\cos B} + \frac{dc}{\cos C} = 0$
- (c) If z is a homogenous function of x and y , $x = e^u + e^{-v}$ and $y = e^{-u} + e^v$ then show that $\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} = x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y}$
- (d) Find approximate error in the surface of rectangular box of sides a , b and c with the error δ is made in the measurement of each side.

OR

Q-5

(6+4+4)

- (a) Find Maximum and Minimum value of following function
 $f(x, y) = x^3 + y^3 - 3x - 12y + 20$
- (b) Find the equations of Tangent plane and Normal line to the surface $z = e^{-x} \sin y$ at point $\left(0, \frac{\pi}{2}\right)$
- (c) If $z^3 - zx - y = 0$ then evaluate $\frac{\partial^2 z}{\partial x \partial y}$

Q-6

(5+4+5)

(a) State modified Euler's theorem and hence find the value of

$$x^2 u_{xx} + 2xy u_{xy} + y^2 u_{yy} \text{ where } u = \tan^{-1} \left(\frac{x+y}{\sqrt{x} + \sqrt{y}} \right)$$

(b) Evaluate: $\lim_{x \rightarrow 1} \frac{x^x - x}{x - 1 - \log x}$

(c) If $y = \cos \{ \log(x^2 + 2x + 1) \}$ then prove that

$$(x+1)^2 y_{n+2} + (2n+1)(x+1)y_{n+1} + (n^2 + 4)y_n = 0$$

OR

Q-6

(4+6+4)

(a) State and prove Euler's theorem on homogeneous function.

(b) Find nth derivative of functions (i) $\frac{x}{(x-1)(2x+3)}$ (ii) $\cos^3 x$

(c) Using Maclaurin's series expansion, expand the function $\cosh x$